Mixed Factoring: Choosing and Combining Methods

Choosing among greatest common factor, trinomial, and special-form methods, and combining more than one technique in a single expression

Directions. Factor each expression *completely.* Nothing tells you in advance which method to use — that decision is the work. Run the checklist below in order on every problem, and expect several expressions to need more than one technique before they are finished.

- **Common factor first.** Pull out the greatest common factor of every term, including any shared variable. Then look at what is left in the bracket and start the checklist again.
- **Count the terms.** Two terms and a subtraction sign point at $A^2 - B^2 = (A - B)(A + B)$. Three terms point at a trinomial.
- **Test for a perfect square** before doing trial pairs: $A^2 \pm 2AB + B^2 = (A \pm B)^2$ needs square ends and a middle term equal to twice the product of their roots.
- **Leading coefficient.** If it is 1, hunt for a factor pair of the constant that adds to the middle coefficient. If it is not, use grouping on the product of the outer coefficients.
- **Complete means complete.** Check every bracket you write to see whether it factors again. A sum of two squares does not.

Show the technique you chose beside each answer, e.g. "GCF, then difference of squares".

1. Factor completely: $6x^2 - 15x$

Answer: ________________

2. Factor completely: $x^2 + 9x + 20$

Answer: ________________

3. Factor completely: $x^2 - 64$

Answer: ________________

4. Factor completely: $3x^2 - 27$

Two terms and a subtraction sign — but check the first step of the checklist before you reach for a pattern.

Answer: ________________

5. Factor completely: $2x^2 + 7x + 3$

Answer: ________________

6. A rectangular flower bed has an area of $x^2 - 3x - 28$ square feet, and each of its side lengths is a whole expression in x.
Factor the area completely to find expressions for the two side lengths.

Answer: ________________

7. Factor completely: $4x^2 - 12x + 9$

Three terms with square ends. Test the middle term before you commit to a perfect square.

Answer: ________________

8. Factor completely: $2x^3 - 18x$

Answer: ________________

9. Factor completely: $6x^2 - 7x - 3$

Answer: ________________

10. A student factored $2x^2 + 8x + 8$ and wrote $(2x + 4)(x + 2)$.
(a) Multiply $(2x + 4)(x + 2)$ out and show what it equals.
(b) Explain in one sentence why the student's answer is still not a complete factorization.
(c) Write the complete factorization of $2x^2 + 8x + 8$ on the answer line below.

Answer: ________________

11. Factor completely: $x^4 - 81$

One of the brackets you write can be factored again. Keep going until no bracket splits over the integers.

Answer: ________________

12. Synthesis challenge. Factor completely: $4x^3 + 4x^2 - 24x$

This one needs two different techniques, used in the right order. Name both.

Answer: ________________